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- **Term:** 2

Laboratory Experiment 006: Pulse-Width Modulation (PWM) and Phase-Locked Loops (PLLs)

Introduction

This notebook demonstrates PWM signal generation and Phase-Locked Loop (PLL) behavior using Python.

Generating a PWM Signal

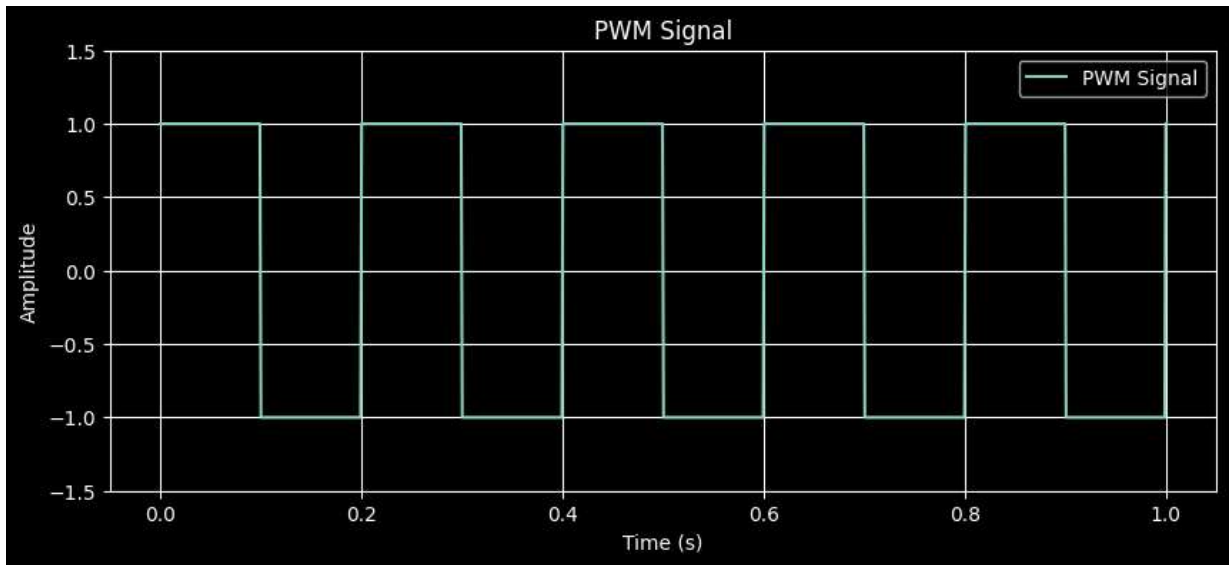
A PWM signal is created by varying the duty cycle of a square wave.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import scipy.signal as signal

# Create a time vector
t = np.linspace(0, 1, 1000)

# Generate a PWM signal
freq = 5 # Hz
duty_cycle = 50 # %
pwm_signal = signal.square(2 * np.pi * freq * t, duty=duty_cycle / 100)

# Plot PWM signal
plt.figure(figsize=(10, 4))
plt.plot(t, pwm_signal, label='PWM Signal')
plt.ylim(-1.5, 1.5)
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('PWM Signal')
plt.legend()
plt.grid()
plt.show()
```



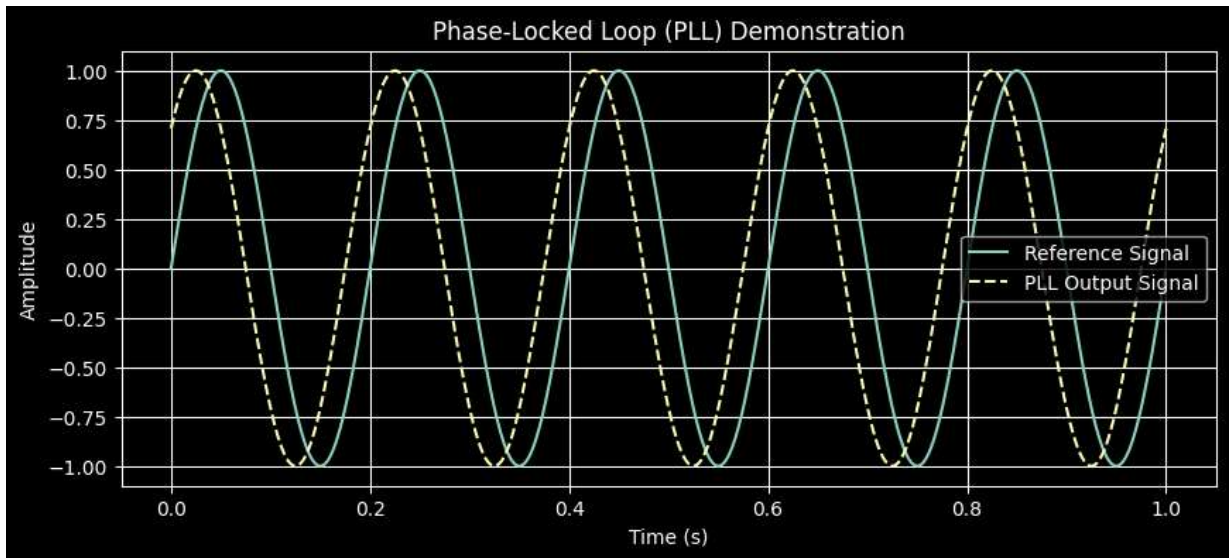
Simulating a Phase-Locked Loop (PLL)

A PLL attempts to synchronize the phase of an output signal to a reference signal.

```
In [2]: # Generate PLL demonstration
# Reference signal
ref_freq = 5 # Hz
ref_signal = np.sin(2 * np.pi * ref_freq * t)

# Simulated PLL output signal (phase-locked)
pll_freq = 5 # Hz
pll_signal = np.sin(2 * np.pi * pll_freq * t + np.pi / 4) # Slight phase shift

# Plot reference and PLL signals
plt.figure(figsize=(10, 4))
plt.plot(t, ref_signal, label='Reference Signal')
plt.plot(t, pll_signal, label='PLL Output Signal', linestyle='dashed')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Phase-Locked Loop (PLL) Demonstration')
plt.legend()
plt.grid()
plt.show()
```

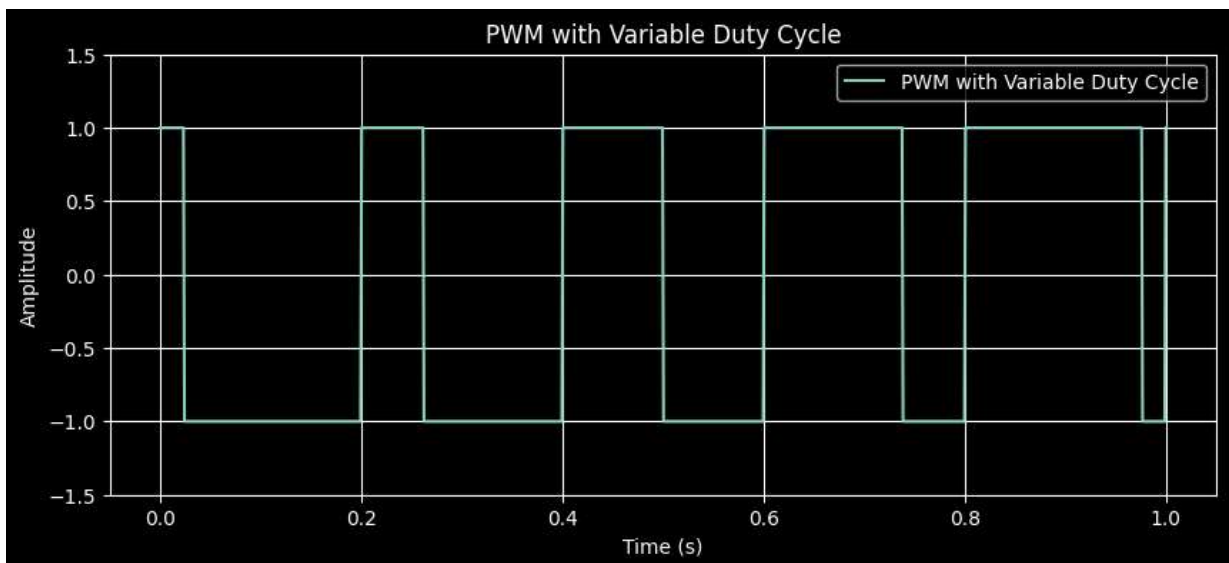


Additional Example: PWM with Variable Duty Cycle

This demonstrates a PWM signal where the duty cycle varies over time.

```
In [3]: # Additional Example: PWM with Variable Duty Cycle
# Define varying duty cycle
duty_cycles = np.linspace(10, 90, len(t))
pwm_variable = signal.square(2 * np.pi * freq * t, duty=duty_cycles / 100)

# Plot PWM with variable duty cycle
plt.figure(figsize=(10, 4))
plt.plot(t, pwm_variable, label='PWM with Variable Duty Cycle')
plt.ylim(-1.5, 1.5)
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('PWM with Variable Duty Cycle')
plt.legend()
plt.grid()
plt.show()
```



Explanation:

- The first plot shows a standard PWM signal with a fixed duty cycle.
- The second plot demonstrates a Phase-Locked Loop (PLL) where the output signal locks onto the reference signal.
- The third plot introduces a PWM signal with a variable duty cycle, which can be used for dynamic power control in applications like motor speed regulation or LED dimming.